Concept 1 | Laws of Exponents and Roots

English Student Edition • Focused formulas and guided practice

Formula opener

Product

$$a^m a^n = a^{m+n}$$

Power

$$(a^m)^n = a^{mn}$$

Root

$$a^{1/n} = \sqrt[n]{a}$$

Eng Eslam Ahmed | 01120009622

Classified Practice **Laws of Exponents and Roots**

Q001 **Written**

Find: (a) 10^0, **(b)** 2^{-3}, **(c)** $(0.2)^{-2}$.

Q002 **Written**

Let $a \neq 0, b \neq 0$. **Simplify: (a)** $a^{-5}a^{8}$, **(b)** $a^{5} \div a^{-3}$, **(c)** $(a^{-1}b)^{2}$, **(d)** $(2^{2})^{3} \div 2^{-2} \times 2^{-3}$.

Classified Practice **Laws of Exponents and Roots**

Q003 **Written**

Find: (a) 5^0, **(b)** 3^{-2}, **(c)** $(0.25)^{-2}$.

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Q004 **Written**

Simplify the eight source expressions: a^5a^4, $a^3 \div a^{-2}$, $(a^{-2}b^4)^{-1}$, $2^3 \times 2^{-7}$, $5^{-4} \div 5^{-6}$, $(a^3b^{-1})^3(a^2b^{-3})^{-2}$, $4^{-5}4^2 \div 4^{-2}$, **and** $2^4 \div (2^{-1})^3 \times 2^{-2}$.

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Q005 **Written**

Find the nine values: $3^0, 11^0, (2/3)^0, 4^{-3}, 5^{-3}, 6^{-2}, (1/3)^{-3}, (3/5)^{-2}, (0.04)^{-2}$.

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Q006 Written

Simplify the twelve same-base expressions in source Question 2: $a^2a^4, a^{-2}a^5, a^{-3}a^{-4}, a^3 \div a^5, a^8 \div a^{-3}, a^{-6} \div a^{-2}, (a^{-1})^3, (a^3b^4)^{-2}, (a^{-3}b^2)^{-3}, 3^5 3^{-3}, 4^4 4^{-3}, 5^5 5^{-5}$.

Classified Practice Laws of Exponents and Roots

Q007 Written

Simplify source Question 3: $5^4 \div 5^2$, $3^2 \div 3^{-2}$, $2^{-4} \div 2^{-5}$, $(a^{-2}b)^3(a^2b^{-1})^2$, $(a^{-3}b)^{-3}(a^4b^2)^{-2}$, $(2a^3b^{-1})^3(-3ab^{-2})^{-3}$, $3^2 3^{-1} \div 3^3$, $5^{-4}5^5 \div 5^{-2}$, $2^4 2^{-8} \div 2^{-6}$, $5^3(5^{-2})^2 \div 5^{-4}$, $3^5 \div (3^{-2})^{-3} \times 3$, **and** $(a/2)^3 a^{-4} \div (2a)^{-2}$.

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Q008 **Written**

Simplify completely with positive exponents: $\left(\frac{2a^2}{b^{-3}}\right)^{-2}(4a^{-3}b^2)^3 \div \left(\frac{a}{2b}\right)^{-4} \div (a^{-2}b^2)$.

Q009 Written

Find $\sqrt[3]{64}$ **and** $\sqrt[3]{1/27}$.

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Q010 **Written**

Perform: $\sqrt[3]{12}\sqrt[3]{18}$, $\sqrt[3]{250}/\sqrt[3]{2}$, $(\sqrt[6]{8})^2$, $\sqrt[3]{\sqrt{729}}$, **and** $\sqrt[3]{54}-\sqrt[3]{16}+\sqrt[3]{2}$.

Q011 Written

Find $\sqrt[4]{81}$ and $\sqrt[4]{1/16}$.

Classified Practice **Laws of Exponents and Roots**

Q012 **Written**

Perform: $\sqrt[4]{8}\sqrt[4]{32}$, $\sqrt[4]{48}/\sqrt[4]{3}$, $(\sqrt[6]{100})^3$, $\sqrt[3]{\sqrt[3]{64}}$, **and** $\sqrt[3]{81} - \sqrt[3]{3} + \sqrt[3]{24}$.

Q013 **Written**

Find: $\sqrt[3]{216}$, $\sqrt[4]{625}$, $\sqrt[5]{243}$, $\sqrt[3]{1/64}$, $\sqrt[3]{1/125}$, $\sqrt[4]{1/81}$, $\sqrt[3]{1}$, $\sqrt[5]{100000}$, $\sqrt[4]{0.0001}$.

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Q014 Written

Perform the fifteen source calculations involving products, quotients, powers and sums of roots (including $\sqrt[3]{2}\sqrt[3]{4}$, $\sqrt[4]{27}\sqrt[4]{243}$, nested roots and the final three like-radical sums).

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Q015 **Written**

Perform $\sqrt[3]{\sqrt[4]{4096}} \div \sqrt{\sqrt[6]{4096}}$.

Q016 **Written**

Express in the form a^r: $\sqrt{5}$, $\sqrt[5]{2^4}$, and $1/\sqrt[5]{3^3}$.

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Q017 **Written**

Find: $27^{2/3}$, $64^{-3/2}$, $(25/4)^{-3/2}$, **and** $[(16/9)^{-3/4}]^{2/3}$.

Q018 **Written**

Express in the form a^r: $\sqrt{a}$, $\sqrt[4]{a^3}$, $\sqrt[6]{a^5}$, $\sqrt[3]{2^2}$, $1/\sqrt{a}$, $1/\sqrt[3]{a^2}$, and $1/\sqrt[4]{5^3}$.

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Q019 **Written**

Find $81^{3/4}$, $16^{-3/2}$, $(25/4)^{-3/2}$, **and** $[(27/125)^{1/2}]^{-4/3}$.

Q020 **Written**

Express the nine source expressions in the form a^r, including roots of a and numerical bases $2, 3, 5$.

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Q021 Written

Evaluate the twelve source expressions with rational powers: $25^{3/2}$, $16^{3/4}$, $64^{2/3}$, $9^{-3/2}$, $125^{-2/3}$, $81^{-3/4}$, $(1/8)^{-1/3}$, $(125/8)^{-2/3}$, $(16/81)^{-3/4}$, [(1/25

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Q022 **Written**

Simplify $\left[\left((1000/27)^{-2/3}\right)^{7/15}\right]^{-15/14}$.

Q023 **Written**

Perform: (a) $2^{2/3}2^{1/2} \div 2^{7/6}$**, (b)** $6^{1/2}36^{1/4}$**, (c)** $\sqrt[3]{6}\sqrt{6}\sqrt[6]{6}$**, (d)** $\sqrt{2} \div \sqrt[6]{32} \times \sqrt[3]{16}$**.**

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Q024 **Written**

Perform: $5^{1/2}5^{1/3} \div 5^{-1/6}$, $4^{5/6}2^{4/3}$, $\sqrt{7}\sqrt[3]{7}\sqrt[6]{7}$, **and** $\sqrt[4]{27}\sqrt{27} \div \sqrt[4]{3}$.

Q025 **Written**

For $a > 0, b > 0$, perform: $\sqrt{a^3} \div \sqrt[3]{a} \div \sqrt[6]{a^5}$ and $\sqrt[3]{a}\sqrt{ab} \div \sqrt[6]{a^2/b^3}$.

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Q026 **Written**

Perform the twelve numerical calculations in source Question 1, including rational powers of 2, 3 and 5 and mixed radical products/quotients.

Q027 Written

For $a > 0, b > 0$, simplify the six source expressions built from $\sqrt{a}, \sqrt[3]{a}, \sqrt[4]{a}, \sqrt[6]{a}$ and mixed a,b radicals.

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Q028 **Written**

Simplify the ministry expression $\left(\frac{(243/32)^{-2/5}(9/4)^{-1}}{(125/64)^{1/3}(0.004)^{-1/2}}\right)^{-3/2}$**.**

Q029 **Written**

Assess the statement $\sqrt{a^2} = a$ for all real a, using positive and negative examples.

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Q030 MCQ

The value of 2^{-3} is:

Choose the correct option.

A 8
B 1/8
C −8
D 1/6

Q031 MCQ

When multiplying equal bases, we:

Choose the correct option.

A subtract exponents

B add exponents

C multiply bases only

D take reciprocals

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Q032 MCQ

The expression $a^5 \div a^{-3}$ simplifies to:

Choose the correct option.

A a^2

B a^8

C a^{-8}

D a^{15}

Q033 **MCQ**

A negative exponent means:

Choose the correct option.

A multiply by -1

B take the reciprocal power

C set the base to zero

D add one

Classified Practice **Laws of Exponents and Roots**

Q034 MCQ

The value of $(0.04)^{-2}$ is:

Choose the correct option.

A 25

B 400

C 625

D 1600

Q035 **MCQ**

The exponent of a in $(a^{-2}b)^3(a^2b^{-1})^2$ is:

Choose the correct option.

A -2

B -1

C 2

D 6

Classified Practice **Laws of Exponents and Roots**

Q036 MCQ

The positive-exponent form of the 6-1 challenge is:

Choose the correct option.

A $a^7 b^6$

B $1/(a^7 b^6)$

C $a^{-7} b^{-6}$

D $1/(a^6 b^7)$

Q037 **MCQ**

The value of $3^5 \div (3^{-2})^{-3} \times 3$ is:

Choose the correct option.

A 1/3

B 1

C 3

D 9

Classified Practice **Laws of Exponents and Roots**

Q038 MCQ

$\sqrt[3]{64}$ equals:

Choose the correct option.

A 2
B 4
C 8
D 16

Q039 **MCQ**

$\sqrt[3]{12}\sqrt[3]{18}$ equals:

Choose the correct option.

A 4

B 5

C 6

D 9

Classified Practice **Laws of Exponents and Roots**

Q040 MCQ

The fourth root of $1/16$ is:

Choose the correct option.

A $1/2$
B $1/4$
C 2
D 4

Q041 **MCQ**

The result of $\sqrt[3]{54} - \sqrt[3]{16} + \sqrt[3]{2}$ is:

Choose the correct option.

A $\sqrt[3]{2}$

B $2\sqrt[3]{2}$

C $3\sqrt[3]{2}$

D 0

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Q042 MCQ

$\sqrt[4]{625}$ **equals:**

Choose the correct option.

A 4

B 5

C 5^2

D 25

Q043 **MCQ**

When combining roots with the same index, multiply:

Choose the correct option.

A the indices

B the radicands

C the exponents only

D the signs only

Classified Practice **Laws of Exponents and Roots**

Q044 MCQ

The roots challenge evaluates to:

Choose the correct option.

A 0
B 1
C 2
D 4

Q045 **MCQ**

$\sqrt[5]{2^4}$ in rational-index form is:

Choose the correct option.

A $2^{4/5}$

B $2^{5/4}$

C 2^{9}

D $4^{1/5}$

Classified Practice **Laws of Exponents and Roots**

Q046 MCQ

The value of $27^{2/3}$ is:

Choose the correct option.

A 3
B 6
C 9
D 18

Q047 MCQ

A reciprocal square root is written as:

Choose the correct option.

A $a^{1/2}$

B $a^{-1/2}$

C a^{-2}

D a^{2}

Classified Practice **Laws of Exponents and Roots**

Q048 MCQ

The value of $81^{-3/4}$ is:

Choose the correct option.

A 27
B 1/9
C 1/27
D 1/81

Q049 **MCQ**

The exponent of a in $\sqrt[4]{a^3}$ is:

Choose the correct option.

A 3/4

B 4/3

C $-3/4$

D 3

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Q050 MCQ

The value of $[(64/27)^{-4/3}]^{-1/2}$ **is:**

Choose the correct option.

A 4/9

B 9/4

C 16/9

D 27/64

Q051 *MCQ*

The 6-3 challenge simplifies to:

Choose the correct option.

A 3/10

B 10/3

C 100/9

D 9/100

Classified Practice **Laws of Exponents and Roots**

Q052 MCQ

$2^{2/3}2^{1/2} \div 2^{7/6}$ **equals:**

Choose the correct option.

A 1/2

B 1

C 2

D 4

Q053 **MCQ**

$4^{5/6}2^{4/3}$ **equals:**

Choose the correct option.

A 4

B 6

C 8

D 16

Classified Practice **Laws of Exponents and Roots**

Q054 MCQ

$\sqrt{a^3} \div \sqrt[3]{a} \div \sqrt[6]{a^5}$ **equals:**

Choose the correct option.

A $a^{1/3}$

B $a^{2/3}$

C $a^{5/3}$

D a

Q055 **MCQ**

The numerical result of $\sqrt[3]{6}\sqrt{6}\sqrt[6]{6}$ is:

Choose the correct option.

A 2

B 3

C 6

D 12

Classified Practice **Laws of Exponents and Roots**

Q056 MCQ

The source expression $\sqrt{a^3} \div \sqrt[3]{a} \div \sqrt[6]{a^5}$ tests:

Choose the correct option.

A integer powers

B combining rational exponents

C logarithms

D trigonometry

Q057 MCQ

The printed decimal in the 6-4 challenge is:

Choose the correct option.

A 0.04

B 0.004

C 0.4

D 4

Classified Practice **Laws of Exponents and Roots**

Q058 **MCQ**

For real a, $\sqrt{a^2}$ equals:

Choose the correct option.

A a always

B $-a$ always

C $|a|$

D 0

Quiz MCQ 1 **Concept Quiz · MCQ**

Which law gives $a^m a^n = a^{m+n}$?

Choose the correct option.

A same-base multiplication

B power of a power

C reciprocal law

D root product

Classified Practice **Laws of Exponents and Roots**

Quiz MCQ 2 **Concept Quiz · MCQ**

The value of $\sqrt[3]{250}/\sqrt[3]{2}$ is:

Choose the correct option.

A 2
B 5
C 10
D 125

Quiz MCQ 3 **Concept Quiz · MCQ**

The value of $64^{-3/2}$ is:

Choose the correct option.

A 512

B 1/64

C 1/512

D 64

Classified Practice **Laws of Exponents and Roots**

Quiz MCQ 4 **Concept Quiz · MCQ**

For a negative real a, $\sqrt{a^2}$ is:

Choose the correct option.

A a

B -a

C 0

D a^2

Quiz WRITTEN 5 **Concept Quiz · Written**

Simplify $a^{-3}a^{7} \div a^{2}$.

Classified Practice **Laws of Exponents and Roots**

Quiz WRITTEN 6 **Concept Quiz · Written**

Simplify $\sqrt[3]{54} - \sqrt[3]{16} + \sqrt[3]{2}$.

Quiz WRITTEN 7 **Concept Quiz · Written**

Evaluate $125^{-2/3}$.

Classified Practice **Laws of Exponents and Roots**

Quiz WRITTEN 8 **Concept Quiz · Written**

Explain why $\sqrt{a^2} = a$ is not true for every real a.

Answer key

Use the question number shown on each page.

Q001 $1, \frac{1}{8}, 25$

Q002 $a^3,\ a^8,\ \frac{b^2}{a^2},\ 32$

Q003 $1, \frac{1}{9}, 16$

Q004 $a^9,\ a^5,\ a^2b^{-4},\ \frac{1}{16},\ 25,\ a^5b^3,\ \frac{1}{4},\ 32$

Q005 $1, 1, 1, \frac{1}{64}, \frac{1}{125}, \frac{1}{36}, 27, \frac{25}{9}, 625$

Q006 $a^6, a^3, a^{-7}, a^{-2}, a^{11}, a^{-4}, a^{-3}, a^{-6}b^{-8}, a^9b^{-6}, 9, 4, 1$

Q007 $25, 81, 2, \frac{b}{a^2}, \frac{a}{b^7}, -\frac{8a^6b^3}{27}, \frac{1}{9}, 125, 4, 125, 1, \frac{a}{2}$

Q008 $\frac{1}{a^7b^6}$

Q009 $4, \frac{1}{3}$

Q010 $6, 5, 2, 3, 2\sqrt[3]{2}$

Q011 $3, \frac{1}{2}$

Q012 $4, 2, 10, \sqrt[3]{4}, 4\sqrt[3]{3}$

Q013 $6, 5, 3, \frac{1}{4}, \frac{1}{5}, \frac{1}{3}, 1, 10, 0.1$

Q014 $2, 4, 9, \sqrt[4]{27}, \frac{1}{4}, \frac{1}{5}, 7, 3, 4, 9, 4, \sqrt[3]{9}, \sqrt[3]{3}, 0, 0$

Q015 1

Q016 $5^{1/2},\ 2^{4/5},\ 3^{-3/5}$

Q017 $9, \frac{1}{512}, \frac{8}{125}, \frac{3}{4}$

Q018 $a^{1/2}, a^{3/4}, a^{5/6}, 2^{2/3}, a^{-1/2}, a^{-2/3}, 5^{-3/4}$

Q019 $27, \frac{1}{64}, \frac{8}{125}, \frac{25}{9}$

Q020 $2^{1/2}, 6^{1/2}, 7^{1/2}, a^{3/4}, a^{5/6}, 2^{2/3}, a^{-1/2}, a^{-2/3}, 5^{-3/4}$

Q021 $125, 8, 16, \frac{1}{27}, \frac{1}{25}, \frac{1}{27}, 2, \frac{4}{25}, \frac{27}{8}, 5, \frac{10}{9}, \frac{16}{9}$

Q022 $\frac{10}{3}$

Q023 $1, 6, 6, 2$

Q024 $5, 8, 7, 9$

Q025 $a^{1/3},\ b\sqrt{a}$

Q026 $1, 9, 5, 9, \sqrt{5}, 4, 5, 3, 2, 1, 8, \sqrt[3]{3}$

Q027 $a^{1/3},\ a^{7/6},\ a^4,\ a^{2/3}b^{5/6},\ a^{1/2}b^3,\ a^{13/6}b^{-5/6}$

Q028 $\left(\frac{2025\sqrt{10}}{64}\right)^{3/2}$

Q029 Sometimes: $\sqrt{a^2} = |a|$; it equals a only when $a \geq 0$.

Q030 B

Q031 B

Q032 B

Q033 B

Q034 C

Q035 A

Q036 B

Q037 B

Q038 B

Q039 C

Q040 A

Q041 B

Q042 B

Q043 B

Q044 B

Q045 A

Q046 C

Q047 B

Q048 C

Q049 A

Q050 C

Q051 B

Q052 B

Q053 C

Q054 A

Q055 C

Q056 B

Q057 B

Q058 C

Quiz MCQ 1 A

Quiz MCQ 2 B

Quiz MCQ 3 C

Quiz MCQ 4 B

Quiz WRITTEN 5 a^2

Quiz WRITTEN 6 $2\sqrt[3]{2}$

Quiz WRITTEN 7 1/25

Quiz WRITTEN 8 $\sqrt{a^2} = |a|$; e.g. $a = -3$ gives $3 \neq -3$